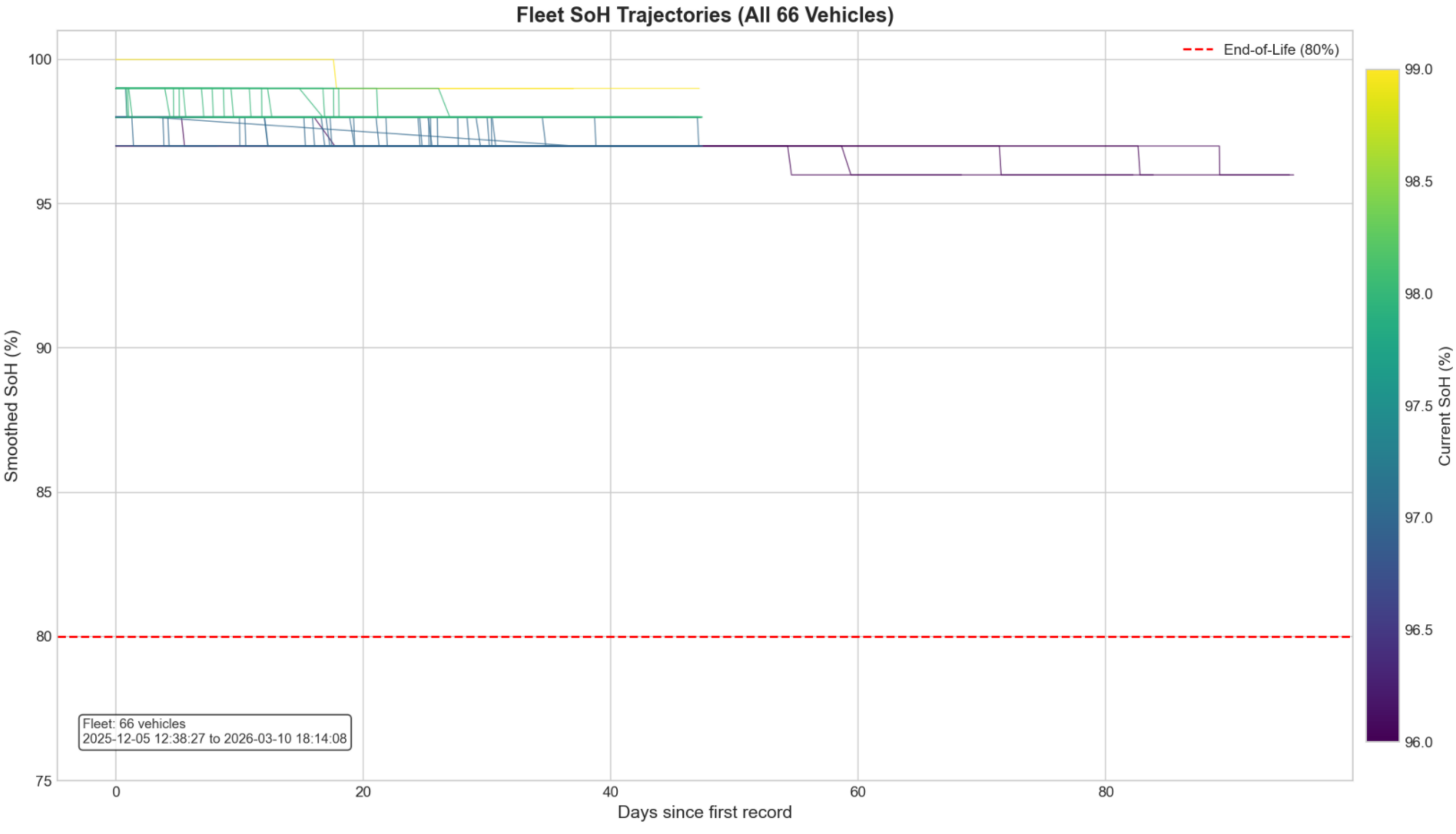
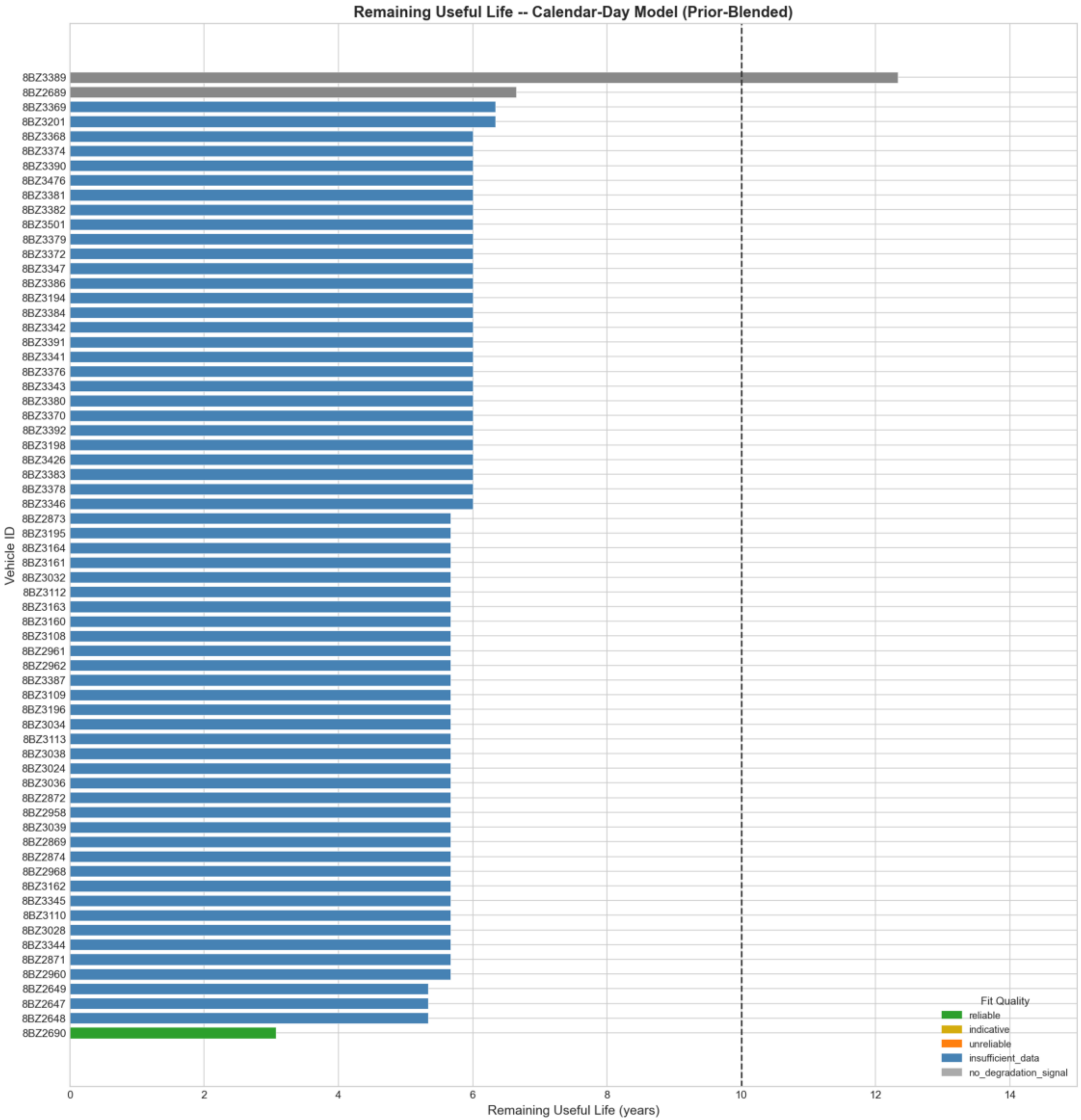

Fleet SoH Trajectories (All 66 Vehicles)
End-of-Life (80%)
Smoothed SoH (%)
100
95
90
85
80
75
0
20
40
60
80
Days since first record
Current SoH (%)
99.0
98.5
98.0
97.5
97.0
96.5
96.0
Fleet: 66 vehicles
2025-12-05 12:38:27 to 2026-03-10 18:14:08


Remaining Useful Life -- Calendar-Day Model (Prior-Blended)
Vehicle ID
8BZ3389
8BZ2689
8BZ3369
8BZ3201
8BZ3368
8BZ3374
8BZ3390
8BZ3476
8BZ3381
8BZ3382
8BZ3501
8BZ3379
8BZ3372
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8BZ3391
8BZ3341
8BZ3376
8BZ3343
8BZ3380
8BZ3370
8BZ3392
8BZ3198
8BZ3426
8BZ3383
8BZ3378
8BZ3346
8BZ2873
8BZ3195
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8BZ3161
8BZ3032
8BZ3112
8BZ3163
8BZ3160
8BZ3108
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8BZ3109
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8BZ3038
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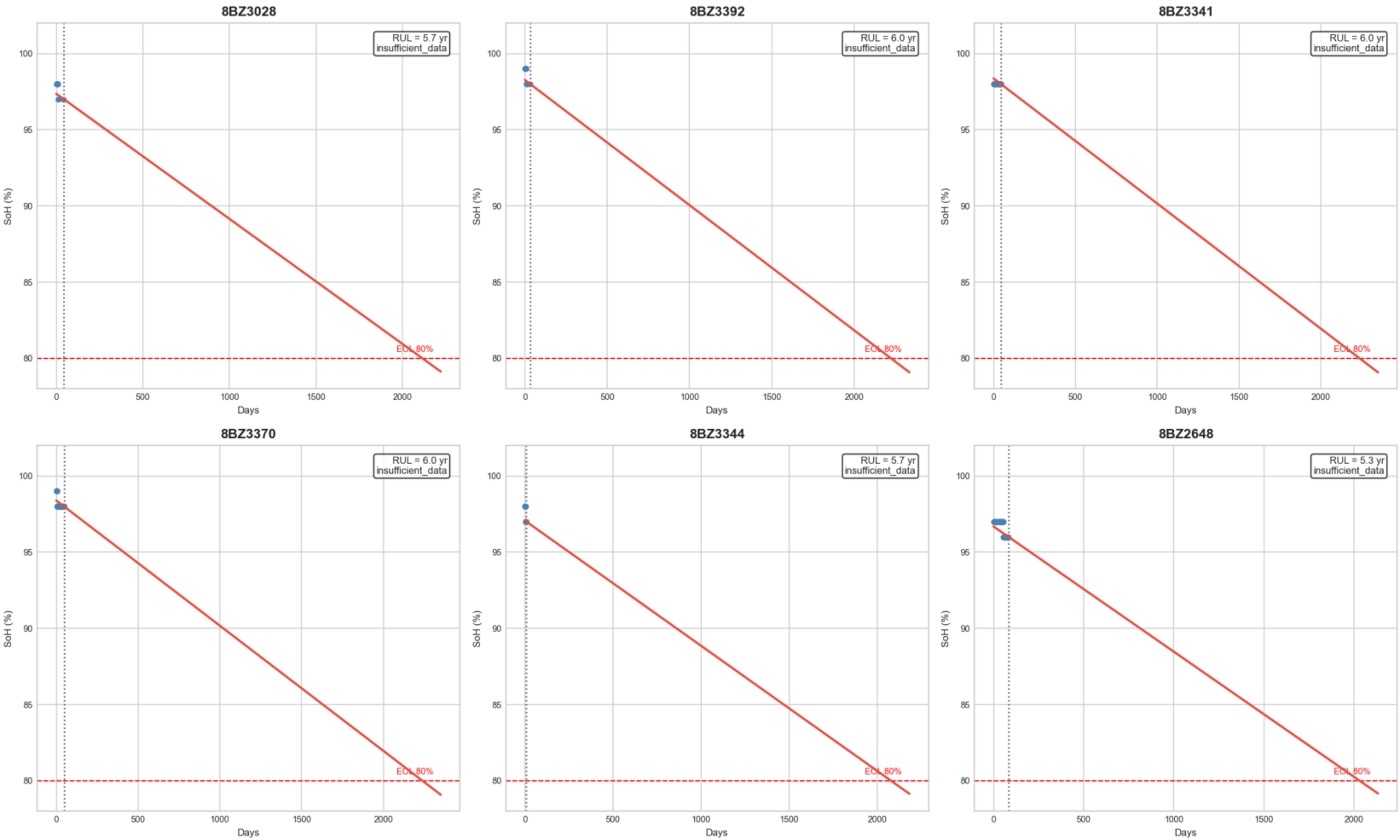
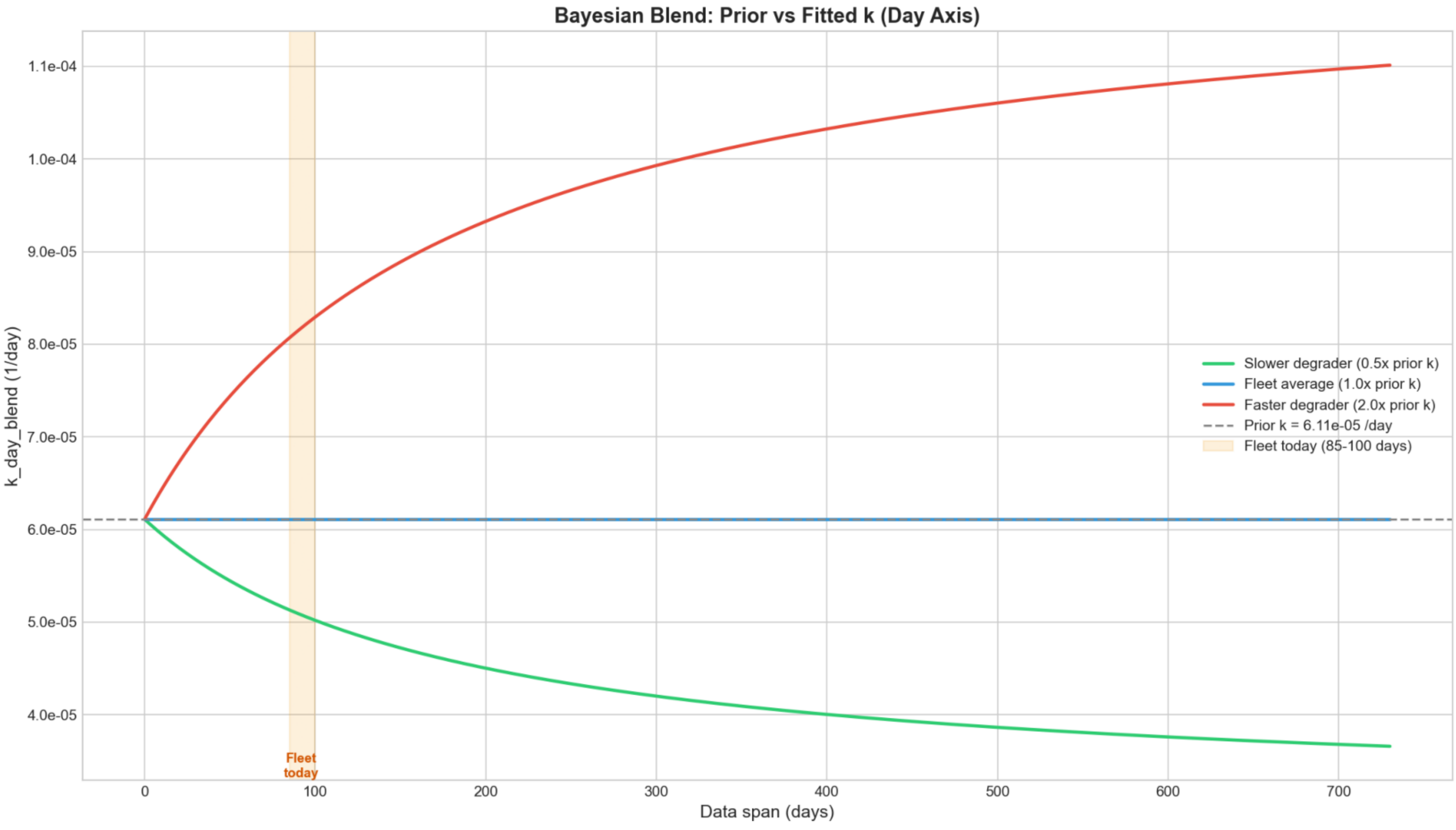
8BZ2874
8BZ2968
8BZ3162
8BZ3345
8BZ3110
8BZ3028
8BZ3344
8BZ2871
8BZ2960
8BZ2649
8BZ2647
8BZ2648
8BZ2690
0
2
4
6
8
10
12
14
Remaining Useful Life (years)
Fit Quality
reliable
indicative
unreliable
insufficient_data
no_degradation_signal

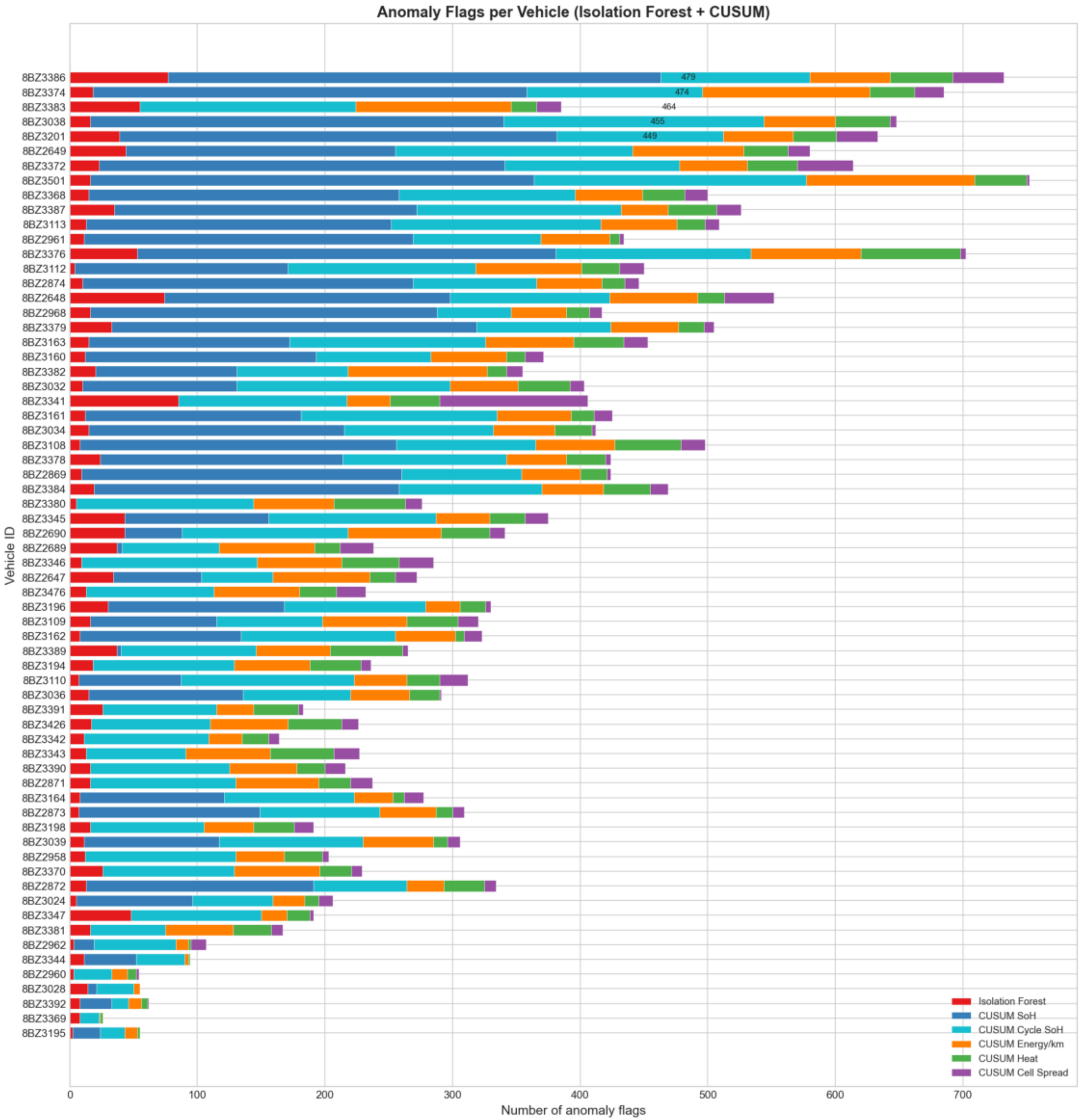
# OLS SoH Trend Fits -- Top-6 Degradation-Signal Vehicles




Bayesian Blend: Prior vs Fitted k (Day Axis)
k_day_blend (1/day)
1.1e-04
1.0e-04
9.0e-05
8.0e-05
7.0e-05
6.0e-05
5.0e-05
4.0e-05
Fleet today
0
100
200
300
400
500
600
700
Data span (days)
Slower degrader (0.5x prior k)
Fleet average (1.0x prior k)
Faster degrader (2.0x prior k)
Prior k = 6.11e-05 /day
Fleet today (85-100 days)

## Degradation Driver Heatmap (Normalised 0-1)

| Vehicle ID | SoH Health (EKF) | Cycle SoH Trend | Voltage Sag | Internal Resistance | Energy/km | Heat Rate | Cell Spread | Composite composite_degradation_score |
|---|---|---|---|---|---|---|---|---|
| 8BZ3028 | 1.00 | 0.02 | 0.34 | 0.76 | 0.41 | 0.78 | 0.83 | 0.65 |
| 8BZ3392 | 0.60 | 1.00 | 0.36 | 0.77 | 0.02 | 1.00 | 0.93 | 0.63 |
| 8BZ3341 | 0.08 | 0.07 | 0.33 | 0.81 | 0.23 | 0.78 | 1.00 | 0.42 |
| 8BZ3370 | 0.39 | 0.03 | 0.40 | 0.85 | 0.19 | 0.70 | 0.23 | 0.41 |
| 8BZ3344 | 0.03 | 0.09 | 1.00 | 1.00 | 0.31 | 0.00 | 0.25 | 0.38 |
| 8BZ2648 | 0.14 | 0.08 | 0.35 | 0.92 | 0.21 | 0.69 | 0.23 | 0.36 |
| 8BZ2958 | 0.11 | 0.07 | 0.35 | 0.86 | 0.19 | 0.78 | 0.31 | 0.36 |
| 8BZ3198 | 0.10 | 0.10 | 0.39 | 0.88 | 0.23 | 0.73 | 0.21 | 0.36 |
| 8BZ3039 | 0.08 | 0.11 | 0.41 | 0.87 | 0.23 | 0.76 | 0.19 | 0.36 |
| 8BZ2871 | 0.08 | 0.06 | 0.34 | 0.91 | 0.18 | 0.70 | 0.35 | 0.35 |
| 8BZ2649 | 0.11 | 0.07 | 0.35 | 0.90 | 0.22 | 0.70 | 0.23 | 0.35 |
| 8BZ3391 | 0.03 | 0.06 | 0.37 | 0.88 | 0.20 | 0.75 | 0.37 | 0.35 |
| 8BZ2960 | 0.05 | 0.07 | 0.36 | 0.82 | 0.23 | 0.76 | 0.38 | 0.35 |
| 8BZ3345 | 0.07 | 0.07 | 0.34 | 0.81 | 0.24 | 0.74 | 0.37 | 0.35 |
| 8BZ3110 | 0.07 | 0.07 | 0.32 | 0.79 | 0.23 | 0.72 | 0.46 | 0.35 |
| 8BZ3162 | 0.07 | 0.07 | 0.38 | 0.84 | 0.23 | 0.71 | 0.31 | 0.35 |
| 8BZ2689 | 0.10 | 0.08 | 0.35 | 0.87 | 0.23 | 0.71 | 0.19 | 0.35 |
| 8BZ3378 | 0.00 | 0.07 | 0.37 | 0.95 | 0.23 | 0.74 | 0.28 | 0.35 |
| 8BZ2647 | 0.11 | 0.08 | 0.35 | 0.89 | 0.17 | 0.70 | 0.23 | 0.34 |
| 8BZ2874 | 0.08 | 0.05 | 0.32 | 0.85 | 0.21 | 0.71 | 0.35 | 0.34 |
| 8BZ2869 | 0.06 | 0.08 | 0.36 | 0.89 | 0.20 | 0.70 | 0.28 | 0.34 |
| 8BZ3389 | 0.07 | 0.08 | 0.38 | 0.76 | 0.29 | 0.80 | 0.20 | 0.34 |
| 8BZ2968 | 0.06 | 0.07 | 0.33 | 0.88 | 0.20 | 0.71 | 0.31 | 0.34 |
| 8BZ2690 | 0.08 | 0.07 | 0.36 | 0.89 | 0.22 | 0.70 | 0.20 | 0.34 |
| 8BZ3201 | 0.00 | 0.06 | 0.36 | 0.81 | 0.27 | 0.76 | 0.38 | 0.34 |
| 8BZ3343 | 0.04 | 0.06 | 0.36 | 0.82 | 0.22 | 0.75 | 0.34 | 0.34 |
| 8BZ3024 | 0.09 | 0.04 | 0.30 | 0.80 | 0.26 | 0.73 | 0.30 | 0.34 |
| 8BZ3038 | 0.06 | 0.06 | 0.35 | 0.87 | 0.19 | 0.71 | 0.29 | 0.34 |
| 8BZ3036 | 0.07 | 0.08 | 0.35 | 0.85 | 0.20 | 0.69 | 0.28 | 0.34 |
| 8BZ3380 | 0.05 | 0.07 | 0.34 | 0.85 | 0.19 | 0.71 | 0.33 | 0.34 |
| 8BZ3113 | 0.07 | 0.06 | 0.35 | 0.81 | 0.22 | 0.71 | 0.31 | 0.34 |
| 8BZ2872 | 0.05 | 0.07 | 0.33 | 0.93 | 0.22 | 0.69 | 0.23 | 0.34 |
| 8BZ3032 | 0.06 | 0.06 | 0.35 | 0.87 | 0.19 | 0.66 | 0.34 | 0.34 |
| 8BZ3374 | 0.05 | 0.06 | 0.40 | 0.95 | 0.05 | 0.69 | 0.32 | 0.34 |
| 8BZ3112 | 0.05 | 0.06 | 0.36 | 0.88 | 0.24 | 0.73 | 0.17 | 0.34 |
| 8BZ3368 | 0.04 | 0.05 | 0.38 | 0.83 | 0.21 | 0.70 | 0.31 | 0.33 |
| 8BZ3376 | 0.03 | 0.08 | 0.38 | 0.76 | 0.26 | 0.77 | 0.27 | 0.33 |
| 8BZ3160 | 0.09 | 0.06 | 0.33 | 0.85 | 0.21 | 0.69 | 0.22 | 0.33 |
| 8BZ3163 | 0.10 | 0.07 | 0.34 | 0.87 | 0.19 | 0.69 | 0.19 | 0.33 |
| 8BZ3108 | 0.05 | 0.06 | 0.35 | 0.84 | 0.22 | 0.71 | 0.25 | 0.33 |
| 8BZ3347 | 0.06 | 0.04 | 0.35 | 0.86 | 0.22 | 0.71 | 0.21 | 0.33 |
| 8BZ3387 | 0.08 | 0.06 | 0.33 | 0.82 | 0.23 | 0.73 | 0.20 | 0.33 |
| 8BZ2962 | 0.01 | 0.07 | 0.39 | 0.65 | 0.21 | 0.77 | 0.48 | 0.33 |
| 8BZ3386 | 0.03 | 0.07 | 0.35 | 0.88 | 0.22 | 0.73 | 0.20 | 0.33 |
| 8BZ3383 | 0.06 | 0.06 | 0.47 | 0.88 | 0.04 | 0.69 | 0.21 | 0.33 |
| 8BZ3426 | 0.05 | 0.05 | 0.34 | 0.84 | 0.19 | 0.71 | 0.30 | 0.33 |
| 8BZ3346 | 0.05 | 0.06 | 0.34 | 0.79 | 0.21 | 0.73 | 0.30 | 0.33 |
| 8BZ2961 | 0.08 | 0.05 | 0.32 | 0.77 | 0.21 | 0.70 | 0.30 | 0.33 |
| 8BZ3379 | 0.04 | 0.07 | 0.35 | 0.83 | 0.27 | 0.76 | 0.14 | 0.33 |
| 8BZ3161 | 0.06 | 0.08 | 0.34 | 0.80 | 0.21 | 0.73 | 0.23 | 0.33 |
| 8BZ3501 | 0.04 | 0.06 | 0.35 | 0.81 | 0.19 | 0.74 | 0.30 | 0.33 |
| 8BZ3034 | 0.08 | 0.07 | 0.32 | 0.82 | 0.23 | 0.71 | 0.16 | 0.33 |
| 8BZ3372 | 0.03 | 0.06 | 0.33 | 0.77 | 0.23 | 0.73 | 0.31 | 0.32 |
| 8BZ3194 | 0.03 | 0.07 | 0.36 | 0.81 | 0.19 | 0.74 | 0.26 | 0.32 |
| 8BZ3384 | 0.05 | 0.05 | 0.37 | 0.80 | 0.21 | 0.75 | 0.18 | 0.32 |
| 8BZ3342 | 0.07 | 0.05 | 0.36 | 0.78 | 0.20 | 0.71 | 0.20 | 0.32 |
| 8BZ3390 | 0.04 | 0.06 | 0.33 | 0.81 | 0.22 | 0.71 | 0.25 | 0.32 |
| 8BZ3109 | 0.03 | 0.06 | 0.35 | 0.87 | 0.15 | 0.65 | 0.29 | 0.32 |
| 8BZ3381 | 0.05 | 0.06 | 0.36 | 0.76 | 0.27 | 0.77 | 0.10 | 0.32 |
| 8BZ3382 | 0.03 | 0.06 | 0.43 | 0.96 | 0.00 | 0.69 | 0.17 | 0.32 |
| 8BZ3196 | 0.06 | 0.07 | 0.33 | 0.81 | 0.20 | 0.65 | 0.24 | 0.32 |
| 8BZ3476 | 0.04 | 0.07 | 0.35 | 0.83 | 0.19 | 0.72 | 0.11 | 0.31 |
| 8BZ3164 | 0.06 | 0.07 | 0.34 | 0.94 | 0.15 | 0.67 | 0.00 | 0.31 |
| 8BZ3369 | 0.07 | 0.00 | 0.18 | 0.70 | 0.54 | 0.57 | 0.22 | 0.31 |
| 8BZ2873 | 0.05 | 0.06 | 0.37 | 0.76 | 0.25 | 0.71 | 0.07 | 0.31 |
| 8BZ3195 | 0.01 | 0.07 | 0.00 | 0.00 | 1.00 | 0.95 | 0.09 | 0.25 |


Anomaly Flags per Vehicle (Isolation Forest + CUSUM)
Vehicle ID
Number of anomaly flags
0
100
200
300
400
500
600
700
479
474
464
455
449
8BZ3386
8BZ3374
8BZ3383
8BZ3038
8BZ3201
8BZ2649
8BZ3372
8BZ3501
8BZ3368
8BZ3387
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8BZ2961
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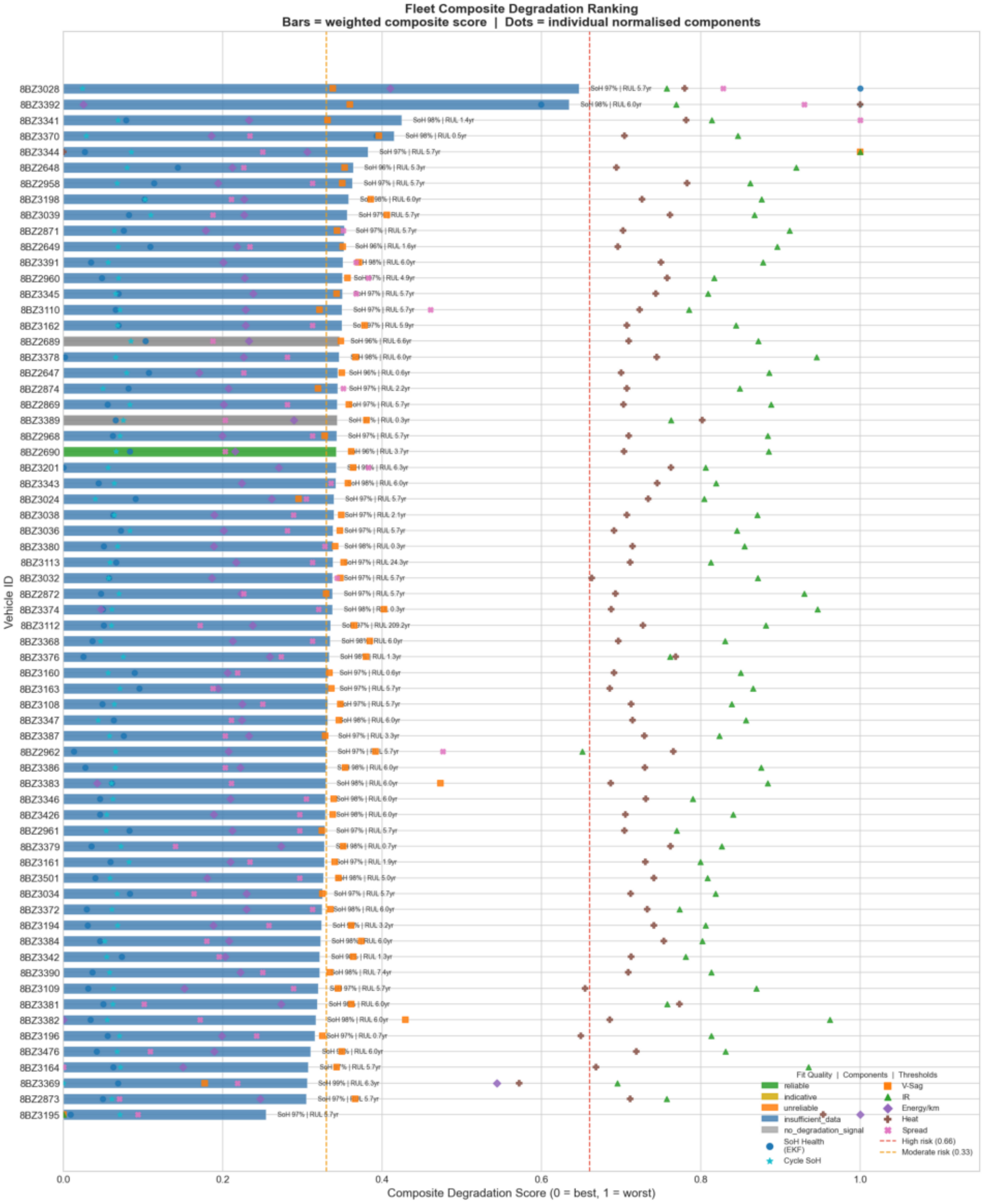
8BZ3347
8BZ3381
8BZ2962
8BZ3344
8BZ2960
8BZ3028
8BZ3392
8BZ3369
8BZ3195
Isolation Forest
CUSUM SoH
CUSUM Cycle SoH
CUSUM Energy/km
CUSUM Heat
CUSUM Cell Spread


Fleet Composite Degradation Ranking
Bars = weighted composite score | Dots = individual normalised components
Vehicle ID
Composite Degradation Score (0 = best, 1 = worst)
0.0
0.2
0.4
0.6
0.8
1.0
8BZ3028
8BZ3392
8BZ3341
8BZ3370
8BZ3344
8BZ2648
8BZ2958
8BZ3198
8BZ3039
8BZ2871
8BZ2649
8BZ3391
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8BZ3378
8BZ2647
8BZ2874
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8BZ2968
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8BZ3201
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8BZ2872
8BZ3374
8BZ3112
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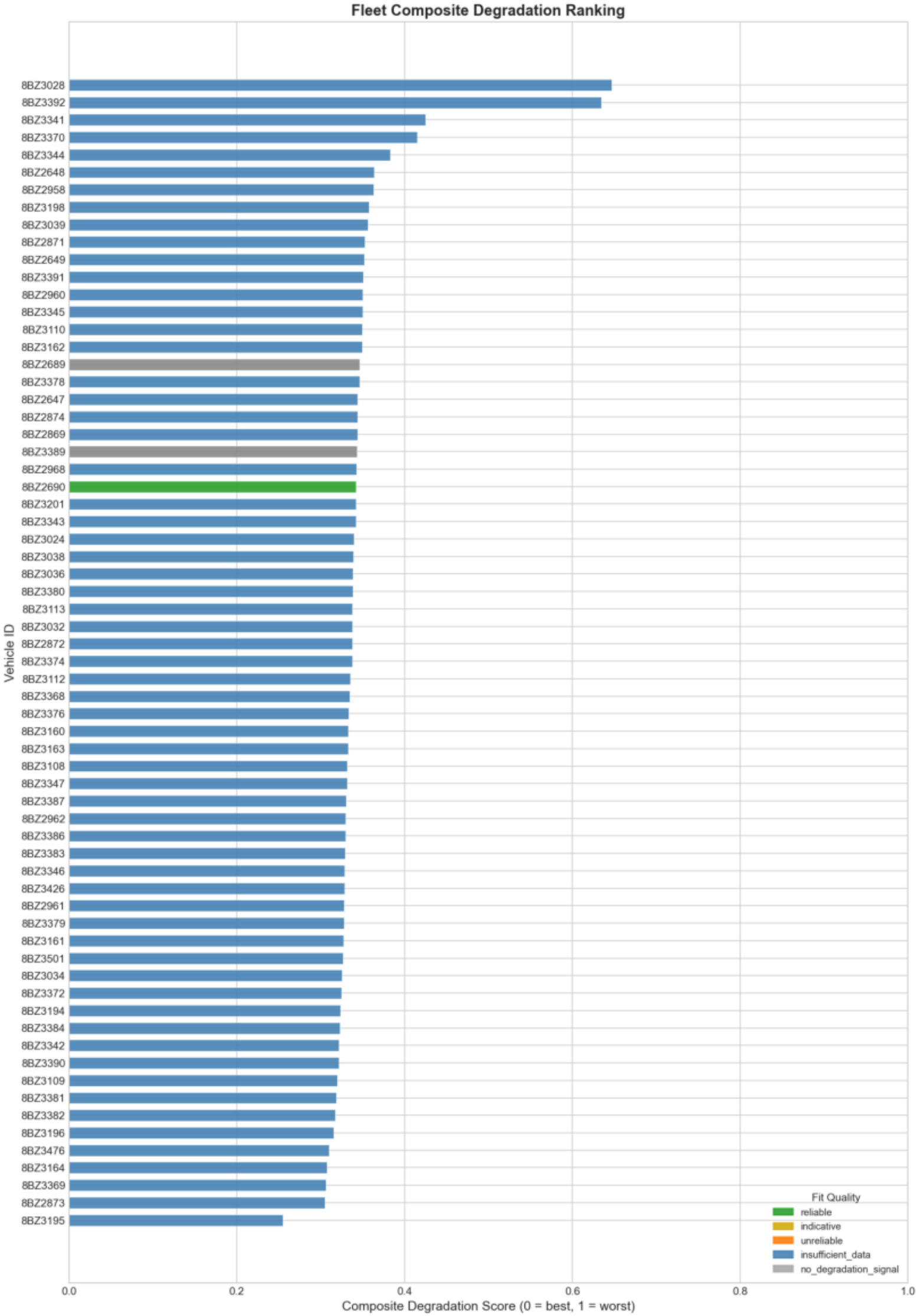
8BZ3109
8BZ3381
8BZ3382
8BZ3196
8BZ3476
8BZ3164
8BZ3369
8BZ2873
8BZ3195
SoH 97% | RUL 5.7yr
SoH 98% | RUL 6.0yr
SoH 98% | RUL 1.4yr
SoH 98% | RUL 0.5yr
SoH 97% | RUL 5.7yr
SoH 96% | RUL 5.3yr
SoH 97% | RUL 5.7yr
SoH 98% | RUL 6.0yr
SoH 97% | RUL 5.7yr
SoH 97% | RUL 5.7yr
SoH 96% | RUL 1.6yr
SoH 98% | RUL 6.0yr
SoH 97% | RUL 4.9yr
SoH 97% | RUL 5.7yr
SoH 97% | RUL 5.7yr
SoH 97% | RUL 5.9yr
SoH 96% | RUL 6.6yr
SoH 96% | RUL 6.0yr
SoH 96% | RUL 0.6yr
SoH 97% | RUL 2.2yr
SoH 97% | RUL 5.7yr
SoH 97% | RUL 0.3yr
SoH 97% | RUL 5.7yr
SoH 96% | RUL 3.7yr
SoH 98% | RUL 6.3yr
SoH 98% | RUL 6.0yr
SoH 97% | RUL 5.7yr
SoH 97% | RUL 2.1yr
SoH 97% | RUL 5.7yr
SoH 98% | RUL 0.3yr
SoH 97% | RUL 24.3yr
SoH 97% | RUL 5.7yr
SoH 97% | RUL 5.7yr
SoH 98% | RUL 0.3yr
SoH 97% | RUL 209.2yr
SoH 98% | RUL 6.0yr
SoH 98% | RUL 1.3yr
SoH 97% | RUL 0.6yr
SoH 97% | RUL 5.7yr
SoH 97% | RUL 5.7yr
SoH 98% | RUL 6.0yr
SoH 97% | RUL 3.3yr
SoH 97% | RUL 5.7yr
SoH 98% | RUL 6.0yr
SoH 98% | RUL 6.0yr
SoH 98% | RUL 6.0yr
SoH 98% | RUL 6.0yr
SoH 97% | RUL 5.7yr
SoH 98% | RUL 0.7yr
SoH 97% | RUL 1.9yr
SoH 98% | RUL 5.0yr
SoH 97% | RUL 5.7yr
SoH 98% | RUL 6.0yr
SoH 98% | RUL 3.2yr
SoH 98% | RUL 6.0yr
SoH 98% | RUL 1.3yr
SoH 98% | RUL 7.4yr
SoH 97% | RUL 5.7yr
SoH 98% | RUL 6.0yr
SoH 98% | RUL 6.0yr
SoH 97% | RUL 0.7yr
SoH 98% | RUL 6.0yr
SoH 97% | RUL 5.7yr
SoH 99% | RUL 6.3yr
SoH 97% | RUL 5.7yr
SoH 97% | RUL 5.7yr
Fit Quality | Components | Thresholds
reliable
indicative
unreliable
insufficient_data
no_degradation_signal
SoH Health (EKF)
Cycle SoH
V-Sag
IR
Energy/km
Heat
Spread
High risk (0.66)
Moderate risk (0.33)


Fleet Composite Degradation Ranking
Vehicle ID
8BZ3028
8BZ3392
8BZ3341
8BZ3370
8BZ3344
8BZ2648
8BZ2958
8BZ3198
8BZ3039
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8BZ2960
8BZ3345
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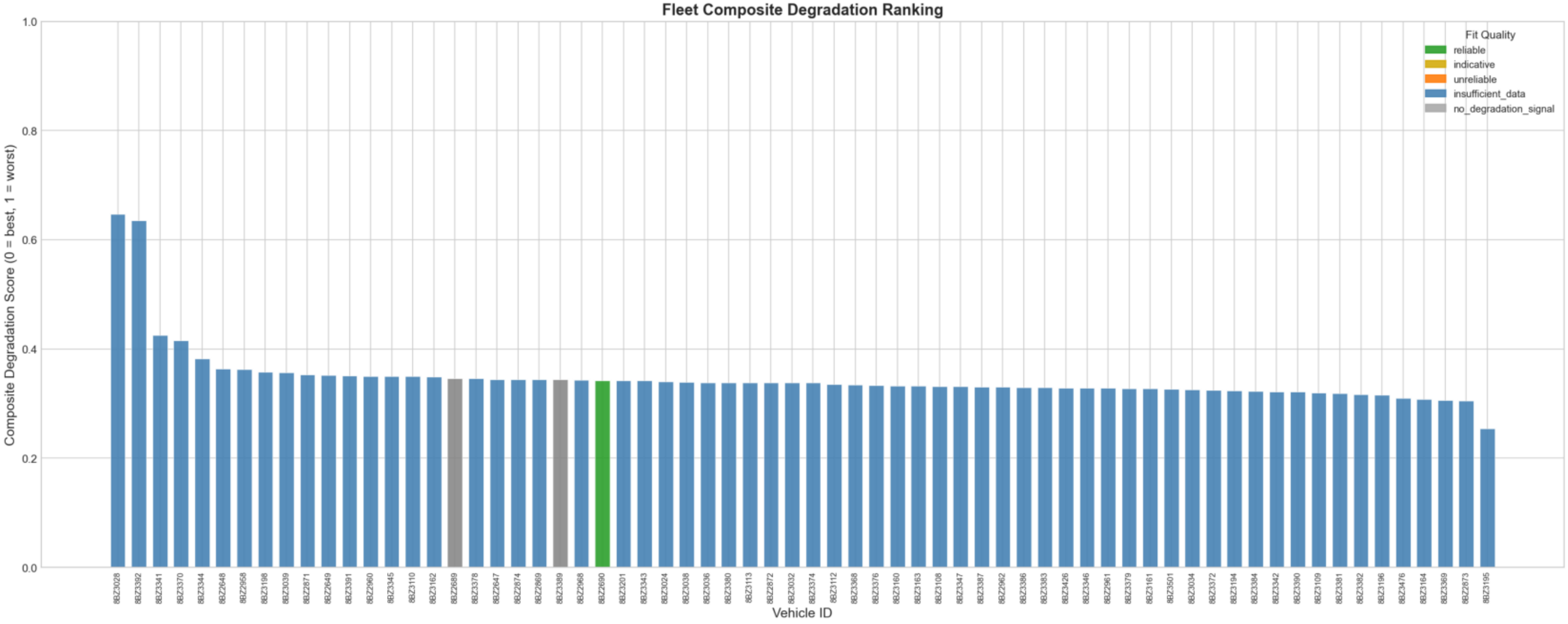
8BZ3390
8BZ3109
8BZ3381
8BZ3382
8BZ3196
8BZ3476
8BZ3164
8BZ3369
8BZ2873
8BZ3195
0.0
0.2
0.4
0.6
0.8
1.0
Composite Degradation Score (0 = best, 1 = worst)
Fit Quality
reliable
indicative
unreliable
insufficient_data
no_degradation_signal


Fleet Composite Degradation Ranking
Composite Degradation Score (0 = best, 1 = worst)
1.0
0.8
0.6
0.4
0.2
0.0
Fit Quality
reliable
indicative
unreliable
insufficient_data
no_degradation_signal
8BZ3028
8BZ3392
8BZ3341
8BZ3370
8BZ3344
8BZ2648
8BZ2958
8BZ3198
8BZ3039
8BZ2871
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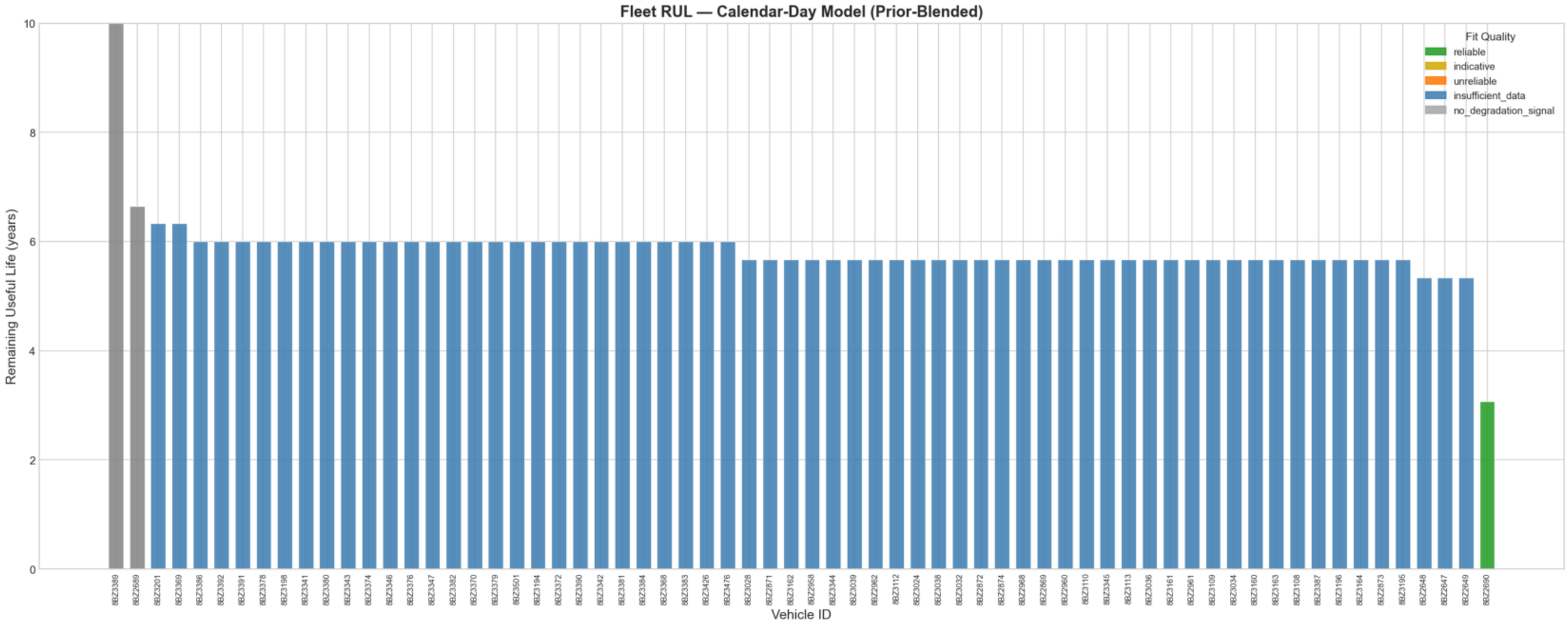
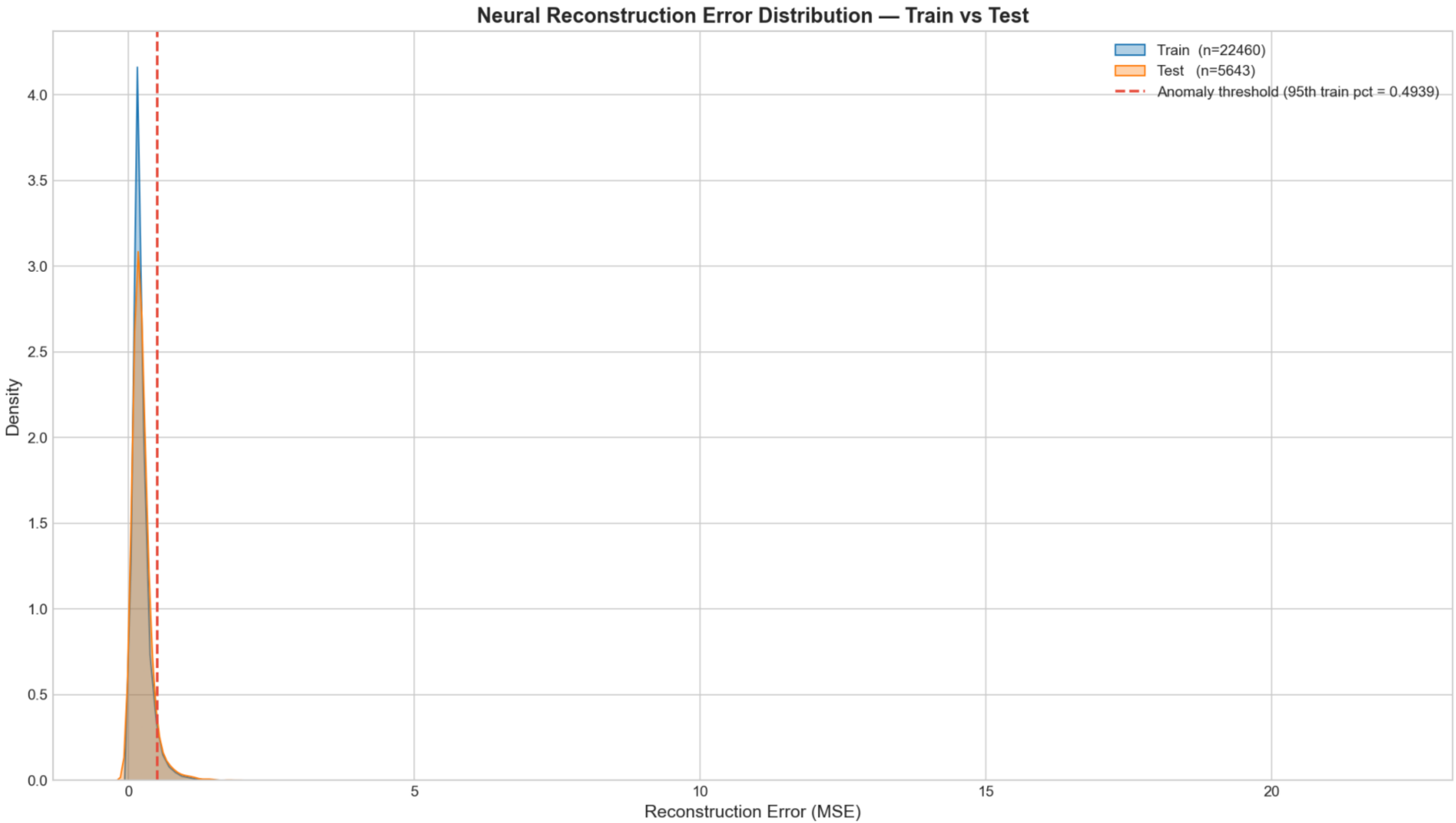
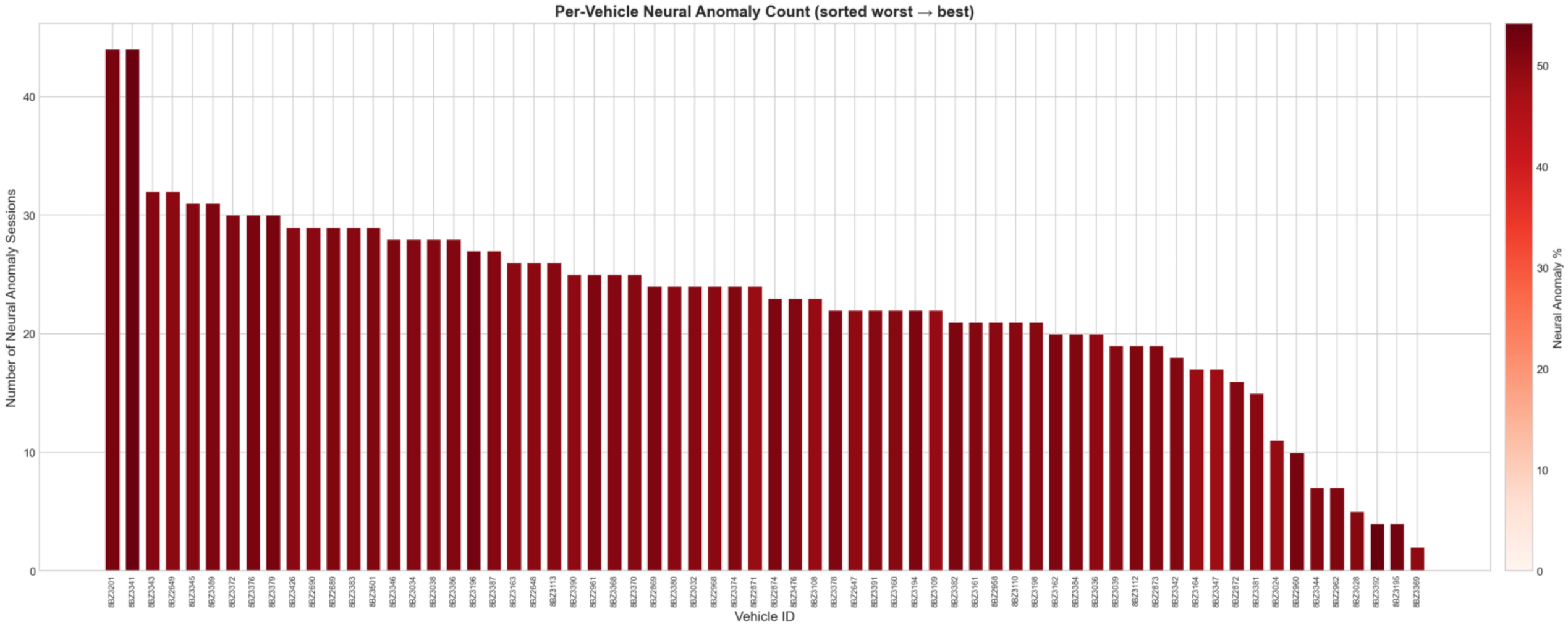
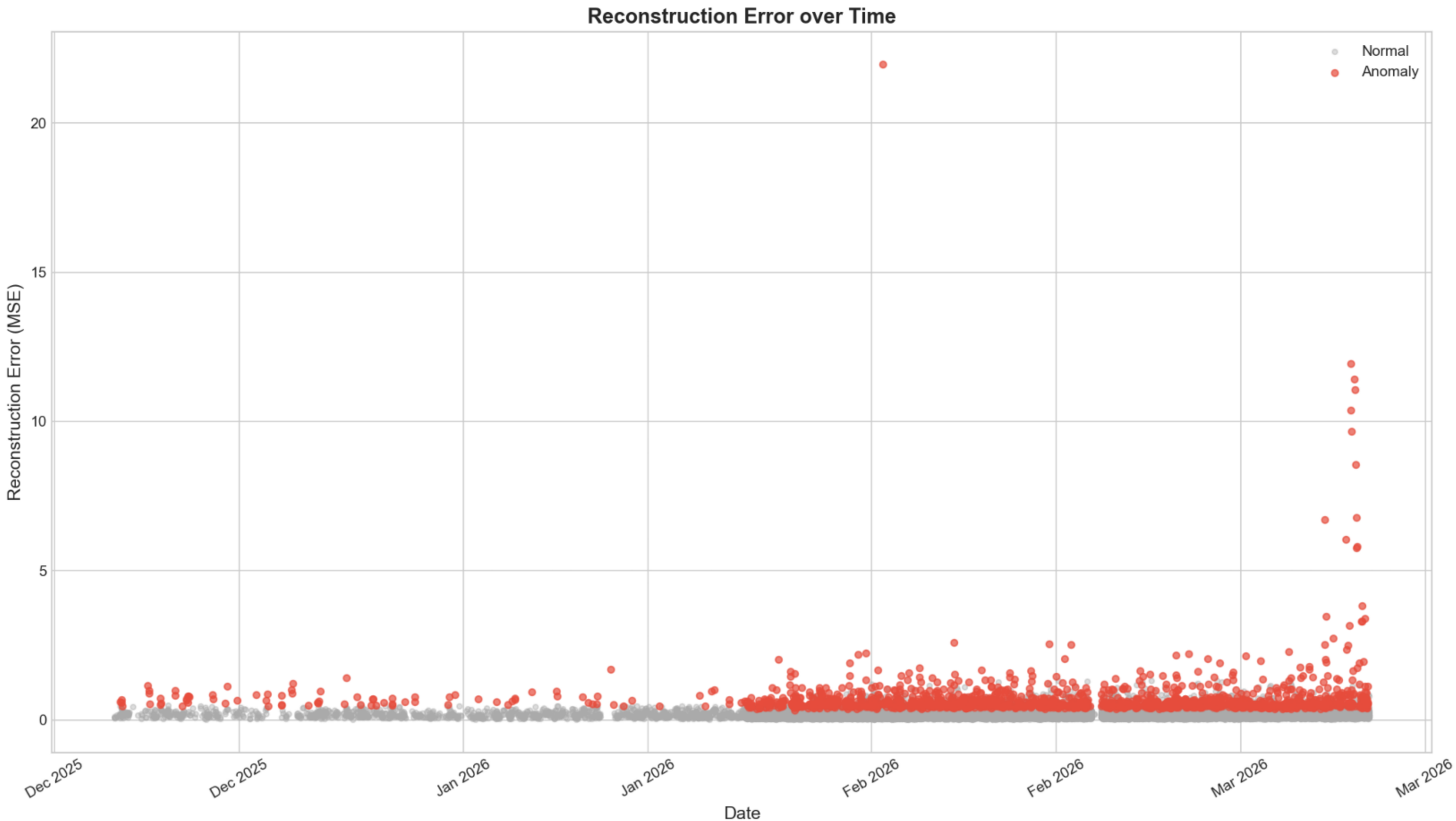
8BZ3195
Vehicle ID

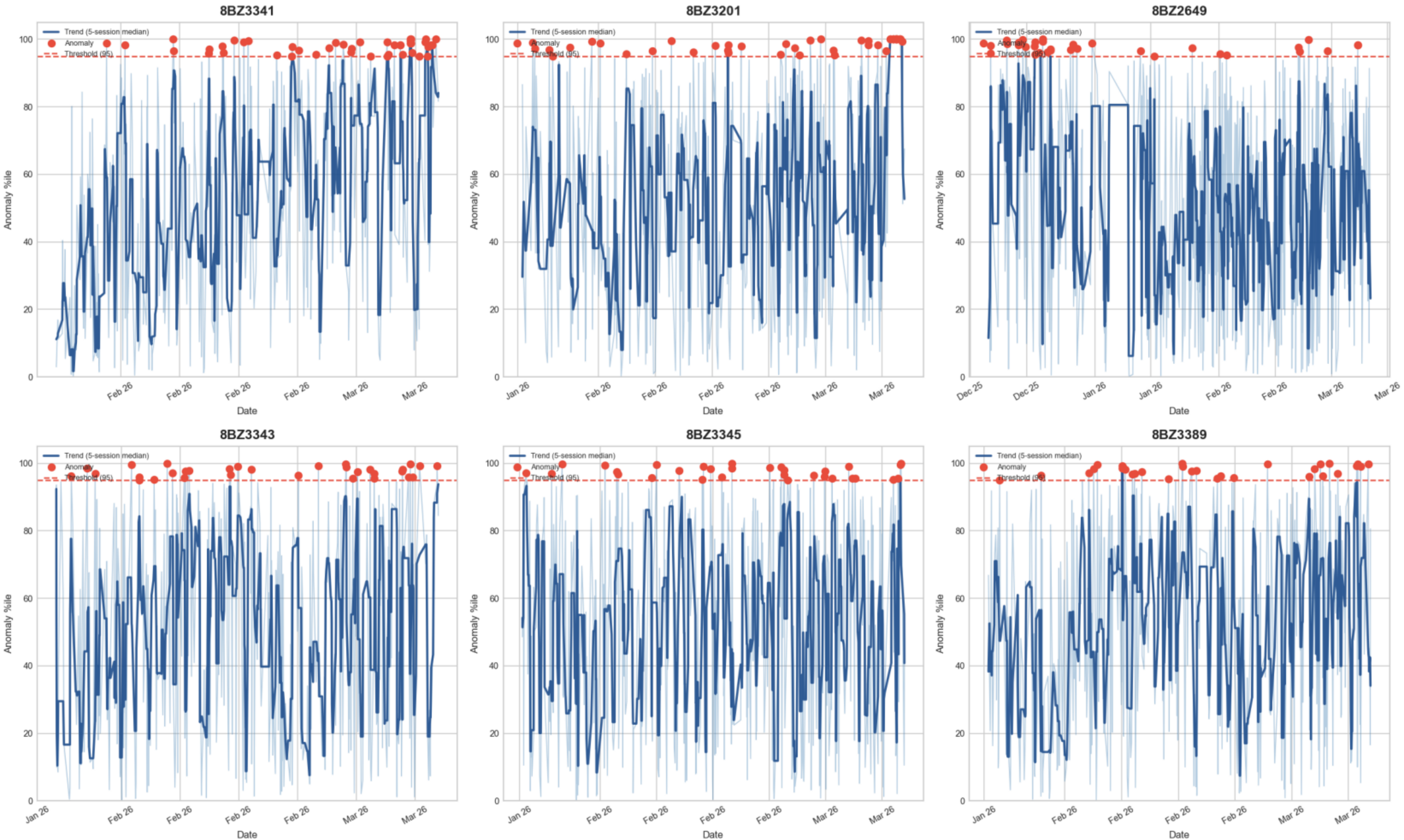

Fleet RUL — Calendar-Day Model (Prior-Blended)
Remaining Useful Life (years)
0
2
4
6
8
10
Vehicle ID
Fit Quality
reliable
indicative
unreliable
insufficient_data
no_degradation_signal
8BZ3389
8BZ2689
8BZ3201
8BZ3369
8BZ3386
8BZ3392
8BZ3391
8BZ3378
8BZ3198
8BZ3341
8BZ3380
8BZ3343
8BZ3374
8BZ3346
8BZ3376
8BZ3347
8BZ3382
8BZ3370
8BZ3379
8BZ3501
8BZ3194
8BZ3372
8BZ3390
8BZ3342
8BZ3381
8BZ3384
8BZ3368
8BZ3383
8BZ3426
8BZ3476
8BZ3028
8BZ2871
8BZ3162
8BZ2958
8BZ3344
8BZ3039
8BZ2962
8BZ3112
8BZ3024
8BZ3038
8BZ3032
8BZ2872
8BZ2874
8BZ2968
8BZ2869
8BZ2960
8BZ3110
8BZ3345
8BZ3113
8BZ3036
8BZ3161
8BZ2961
8BZ3109
8BZ3034
8BZ3160
8BZ3163
8BZ3108
8BZ3387
8BZ3196
8BZ3164
8BZ2873
8BZ3195
8BZ2648
8BZ2647
8BZ2649
8BZ2690

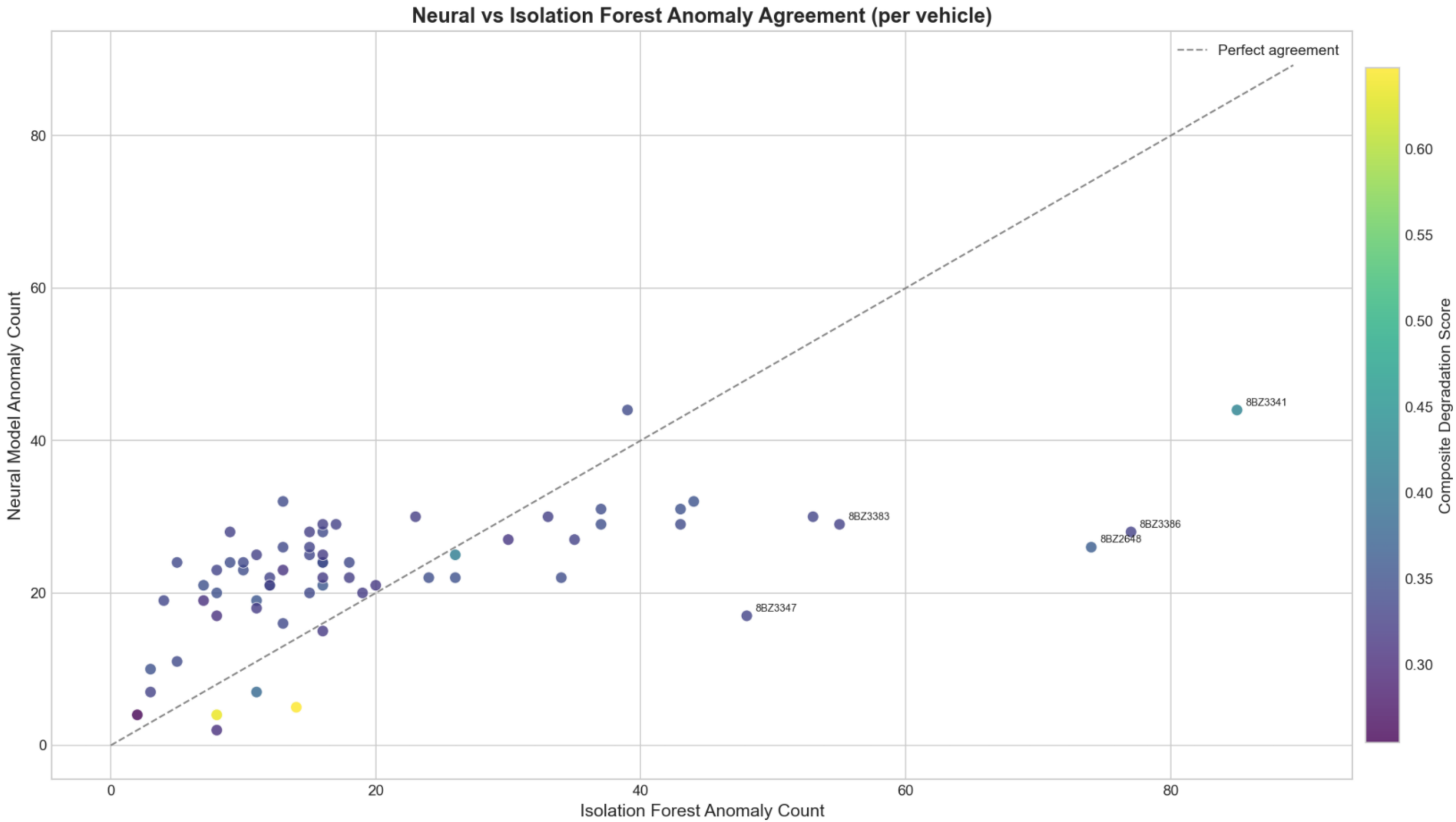

Neural Reconstruction Error Distribution — Train vs Test
Train (n=22460)
Test (n=5643)
Anomaly threshold (95th train pct = 0.4939)
Density
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
5
10
15
20
Reconstruction Error (MSE)

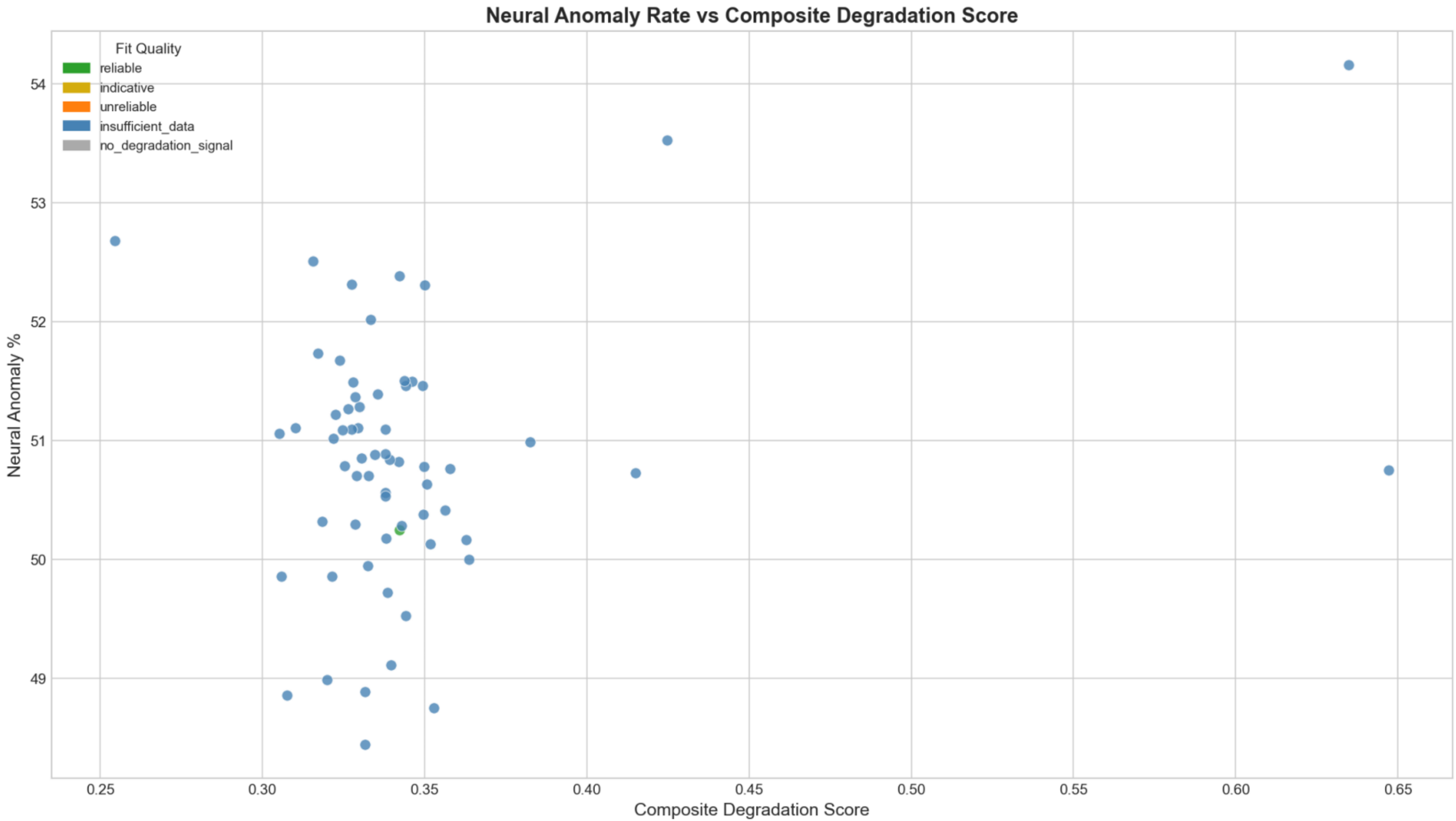

Per-Vehicle Neural Anomaly Count (sorted worst → best)
Number of Neural Anomaly Sessions
0
10
20
30
40
Vehicle ID
8BZ3201
8BZ3341
8BZ3343
8BZ2649
8BZ3345
8BZ3389
8BZ3372
8BZ3376
8BZ3379
8BZ3426
8BZ2690
8BZ2689
8BZ3383
8BZ3501
8BZ3346
8BZ3034
8BZ3038
8BZ3386
8BZ3196
8BZ3387
8BZ3163
8BZ2648
8BZ3113
8BZ3390
8BZ2961
8BZ3368
8BZ3370
8BZ2869
8BZ3380
8BZ3032
8BZ2968
8BZ3374
8BZ2871
8BZ2874
8BZ3476
8BZ3108
8BZ3378
8BZ2647
8BZ3391
8BZ3160
8BZ3194
8BZ3109
8BZ3382
8BZ3161
8BZ2958
8BZ3110
8BZ3198
8BZ3162
8BZ3384
8BZ3036
8BZ3039
8BZ3112
8BZ2873
8BZ3342
8BZ3164
8BZ3347
8BZ2872
8BZ3381
8BZ3024
8BZ2960
8BZ3344
8BZ2962
8BZ3028
8BZ3392
8BZ3195
8BZ3369
Neural Anomaly %
0
10
20
30
40
50

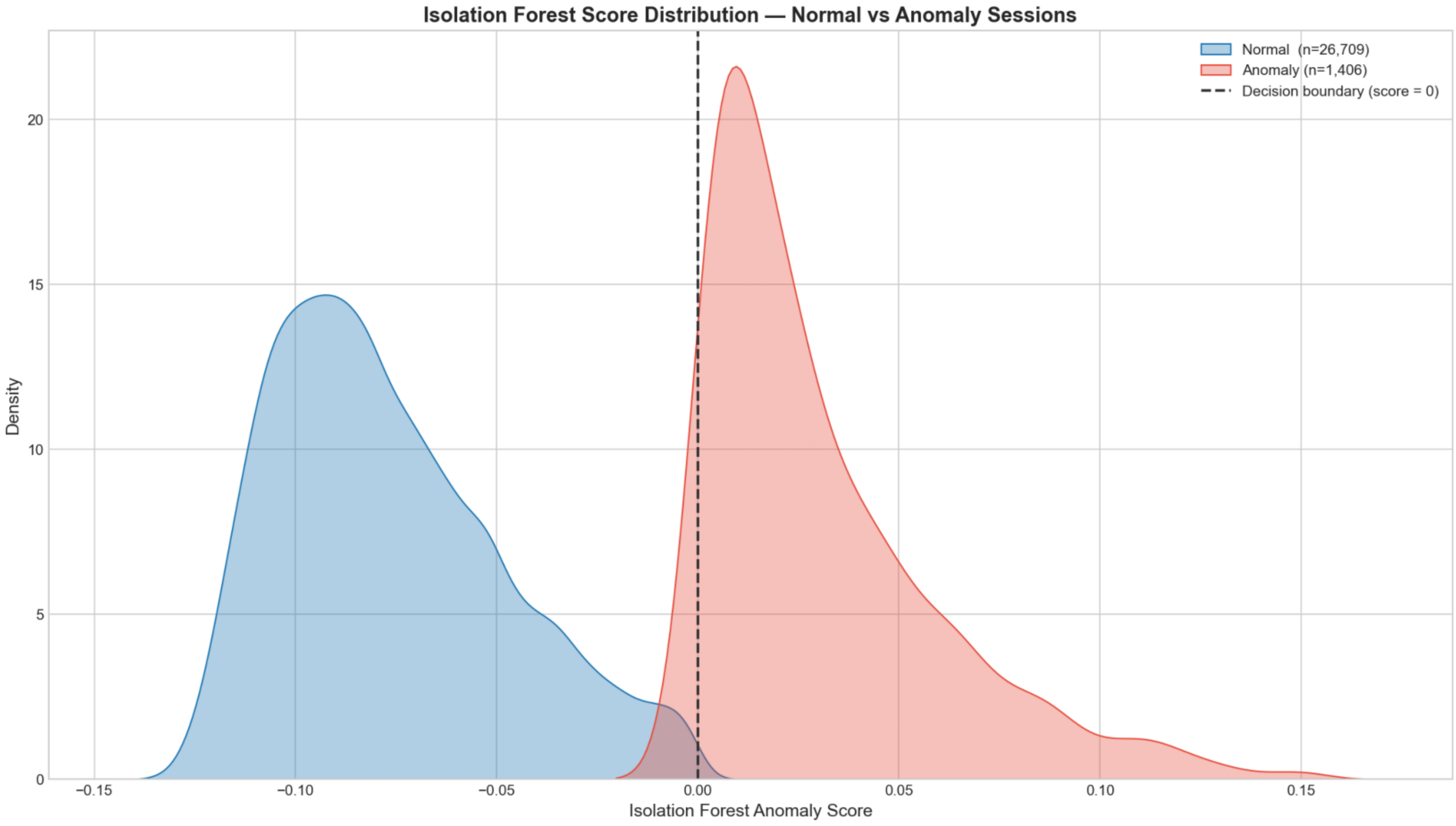

Reconstruction Error over Time
Normal
Anomaly
Reconstruction Error (MSE)
20
15
10
5
0
Dec 2025
Dec 2025
Jan 2026
Jan 2026
Feb 2026
Feb 2026
Mar 2026
Mar 2026
Date

# Neural Anomaly Score Over Time — Top-6 Vehicles (by anomaly count)




Neural vs Isolation Forest Anomaly Agreement (per vehicle)
Perfect agreement
Neural Model Anomaly Count
Isolation Forest Anomaly Count
Composite Degradation Score
0
20
40
60
80
0.30
0.35
0.40
0.45
0.50
0.55
0.60
8BZ3341
8BZ3383
8BZ3386
8BZ2648
8BZ3347


Neural Anomaly Rate vs Composite Degradation Score
Fit Quality
reliable
indicative
unreliable
insufficient_data
no_degradation_signal
Neural Anomaly %
54
53
52
51
50
49
0.25
0.30
0.35
0.40
0.45
0.50
0.55
0.60
0.65
Composite Degradation Score

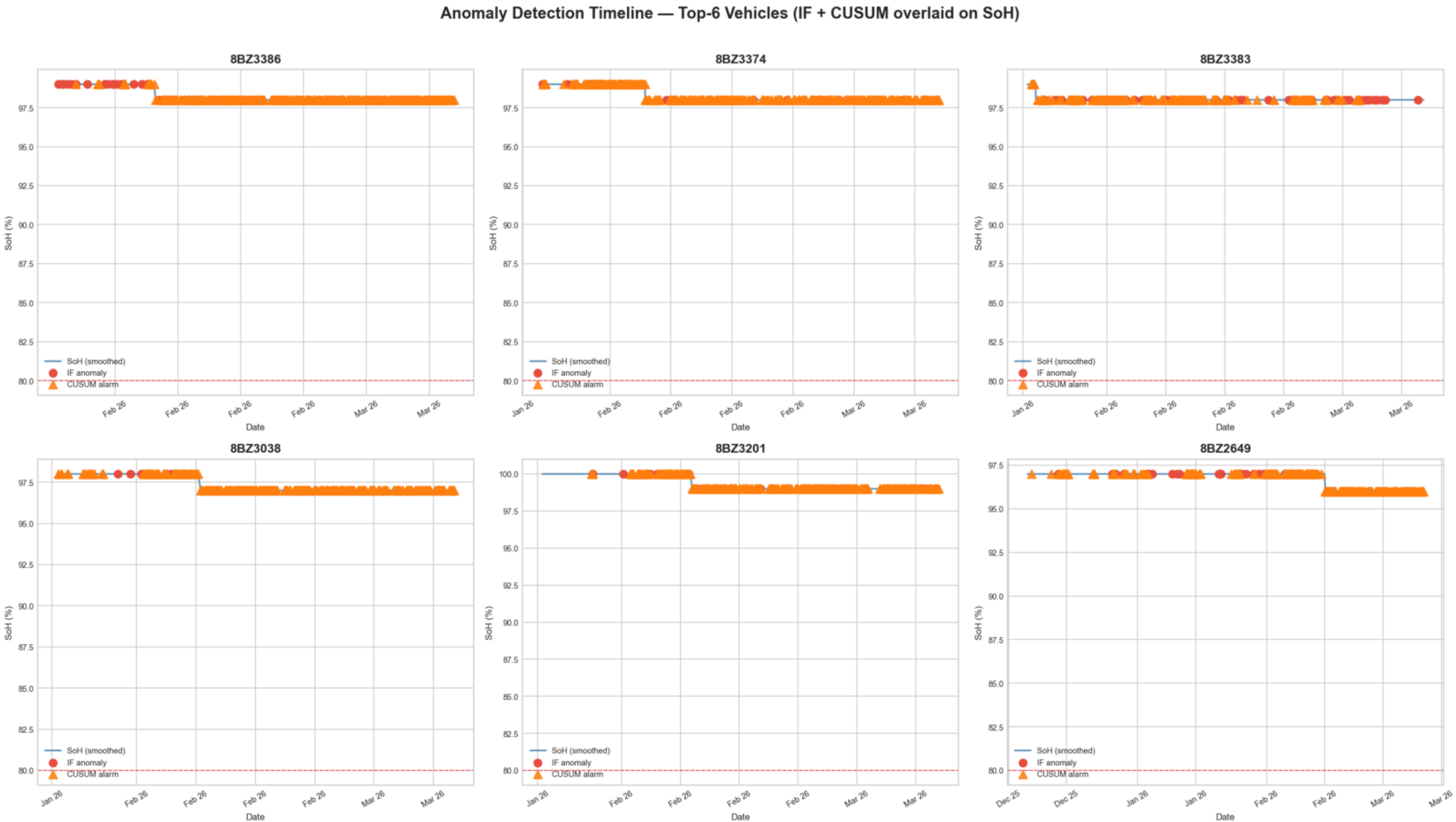

Isolation Forest Score Distribution — Normal vs Anomaly Sessions
Normal (n=26,709)
Anomaly (n=1,406)
Decision boundary (score = 0)
Density
0
5
10
15
20
−0.15
−0.10
−0.05
0.00
0.05
0.10
0.15
Isolation Forest Anomaly Score

## CUSUM Alarm Counts per Vehicle × Alarm Type
(vehicles with ≥1 alarm, sorted by total)

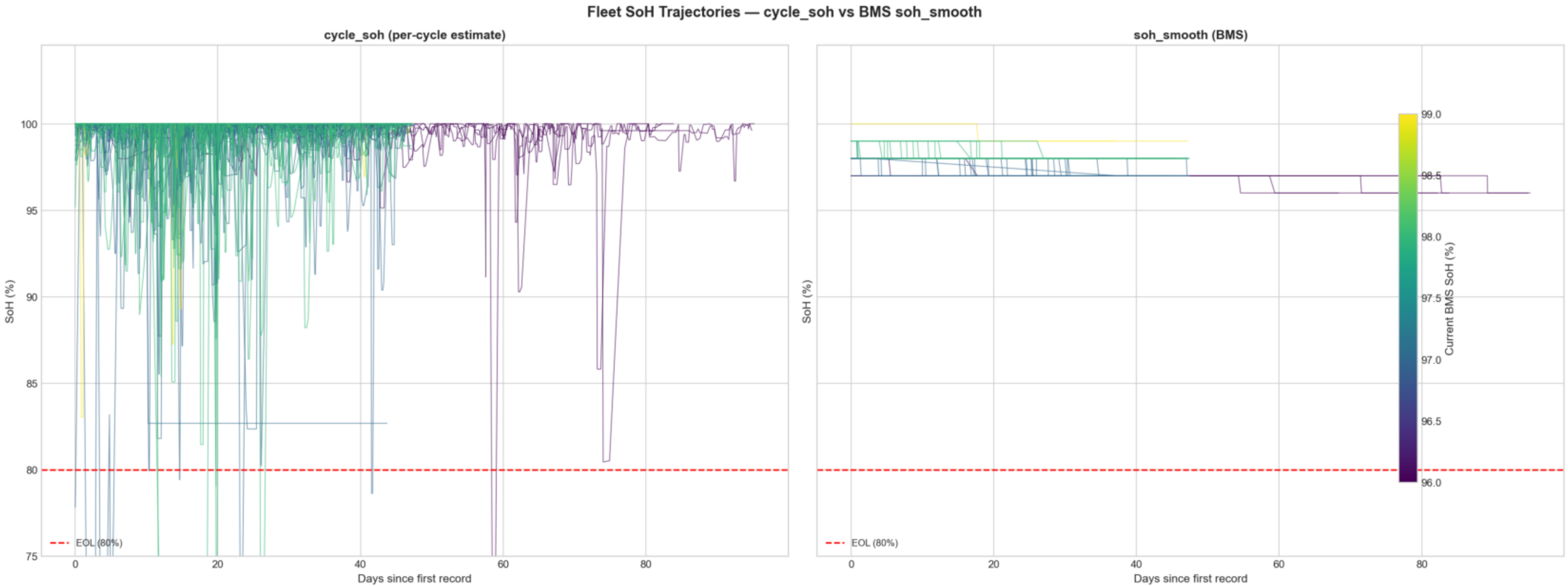
| Vehicle ID | SoH (BMS) | Cycle SoH | Energy/km | Heat Rate | Cell Spread |
|---|---|---|---|---|---|
| 8BZ3501 | 348 | 213 | 132 | 41 | 2 |
| 8BZ3374 | 340 | 138 | 131 | 35 | 23 |
| 8BZ3386 | 386 | 117 | 63 | 49 | 40 |
| 8BZ3376 | 328 | 153 | 86 | 78 | 4 |
| 8BZ3038 | 324 | 204 | 56 | 43 | 5 |
| 8BZ3201 | 343 | 130 | 55 | 34 | 32 |
| 8BZ3372 | 318 | 137 | 53 | 39 | 44 |
| 8BZ2649 | 211 | 186 | 87 | 35 | 17 |
| 8BZ3113 | 239 | 164 | 60 | 22 | 11 |
| 8BZ3387 | 237 | 160 | 37 | 38 | 19 |
| 8BZ3108 | 248 | 109 | 62 | 52 | 19 |
| 8BZ3368 | 243 | 138 | 53 | 33 | 18 |
| 8BZ2648 | 224 | 125 | 69 | 21 | 39 |
| 8BZ3379 | 286 | 105 | 53 | 20 | 8 |
| 8BZ3384 | 239 | 112 | 48 | 37 | 14 |
| 8BZ3112 | 167 | 147 | 83 | 30 | 19 |
| 8BZ3163 | 157 | 154 | 69 | 39 | 19 |
| 8BZ2874 | 259 | 97 | 51 | 18 | 11 |
| 8BZ2961 | 258 | 100 | 54 | 8 | 3 |
| 8BZ2869 | 251 | 94 | 46 | 21 | 3 |
| 8BZ3161 | 169 | 154 | 58 | 18 | 14 |
| 8BZ2968 | 272 | 58 | 43 | 18 | 10 |
| 8BZ3378 | 190 | 128 | 47 | 31 | 4 |
| 8BZ3034 | 200 | 117 | 48 | 29 | 3 |
| 8BZ3032 | 121 | 167 | 53 | 41 | 11 |
| 8BZ3160 | 181 | 90 | 59 | 15 | 14 |
| 8BZ3382 | 111 | 87 | 109 | 15 | 13 |
| 8BZ3345 | 113 | 131 | 42 | 28 | 18 |
| 8BZ3383 | 0 | 169 | 122 | 20 | 19 |
| 8BZ3341 | 0 | 132 | 34 | 39 | 116 |
| 8BZ2872 | 178 | 73 | 29 | 32 | 9 |
| 8BZ3162 | 126 | 121 | 47 | 7 | 14 |
| 8BZ3110 | 80 | 136 | 41 | 26 | 22 |
| 8BZ3109 | 99 | 83 | 66 | 40 | 16 |
| 8BZ2873 | 142 | 94 | 44 | 13 | 9 |
| 8BZ3196 | 138 | 111 | 27 | 20 | 4 |
| 8BZ2690 | 45 | 130 | 73 | 38 | 12 |
| 8BZ3039 | 106 | 113 | 55 | 11 | 10 |
| 8BZ3346 | 0 | 138 | 66 | 45 | 27 |
| 8BZ3036 | 121 | 84 | 46 | 24 | 1 |
| 8BZ3380 | 0 | 139 | 63 | 56 | 13 |
| 8BZ3164 | 113 | 102 | 30 | 9 | 15 |
| 8BZ2647 | 69 | 56 | 76 | 20 | 17 |
| 8BZ3389 | 3 | 106 | 58 | 57 | 4 |
| 8BZ2871 | 0 | 114 | 65 | 25 | 17 |
| 8BZ3476 | 0 | 100 | 67 | 29 | 23 |
| 8BZ3194 | 0 | 111 | 59 | 40 | 8 |
| 8BZ3343 | 0 | 78 | 66 | 50 | 20 |
| 8BZ3426 | 0 | 93 | 61 | 42 | 13 |
| 8BZ3370 | 0 | 103 | 67 | 25 | 8 |
| 8BZ2689 | 4 | 76 | 75 | 20 | 26 |
| 8BZ3024 | 91 | 63 | 25 | 11 | 11 |
| 8BZ3390 | 0 | 109 | 53 | 22 | 16 |
| 8BZ2958 | 0 | 118 | 38 | 30 | 5 |
| 8BZ3198 | 0 | 89 | 39 | 32 | 15 |
| 8BZ3391 | 0 | 89 | 29 | 35 | 4 |
| 8BZ3342 | 0 | 98 | 26 | 21 | 8 |
| 8BZ3381 | 0 | 59 | 53 | 30 | 9 |
| 8BZ3347 | 0 | 102 | 20 | 18 | 3 |
| 8BZ2962 | 16 | 64 | 10 | 2 | 12 |
| 8BZ3344 | 41 | 38 | 3 | 1 | 0 |
| 8BZ3392 | 25 | 13 | 10 | 5 | 1 |
| 8BZ3195 | 22 | 19 | 10 | 2 | 0 |
| 8BZ2960 | 0 | 30 | 12 | 7 | 2 |
| 8BZ3028 | 7 | 29 | 5 | 0 | 0 |
| 8BZ3369 | 0 | 15 | 1 | 2 | 0 |

CUSUM Channel

Alarm count: 0, 50, 100, 150, 200, 250, 300, 350


Anomaly Detection Timeline — Top-6 Vehicles (IF + CUSUM overlaid on SoH)
8BZ3386
8BZ3374
8BZ3383
8BZ3038
8BZ3201
8BZ2649
SoH (%)
Date
SoH (smoothed)
IF anomaly
CUSUM alarm
Jan 26
Feb 26
Mar 26
Dec 25

# EKF SoH Tracking — Top Vehicles by Degradation Score
(green trace = filtered estimate, band = ±1σ, dots = observed capacity_soh)


Fleet SoH Trajectories — cycle_soh vs BMS soh_smooth
cycle_soh (per-cycle estimate)
soh_smooth (BMS)
SoH (%)
Days since first record
EOL (80%)
Current BMS SoH (%)
100
95
90
85
80
75
0
20
40
60
80
99.0
98.5
98.0
97.5
97.0
96.5
96.0

# cycle_soh vs EKF SoH — Top Vehicles by Degradation Score
(green = EKF filtered, cyan triangles = cycle_soh observations)